

Dimensional regularisation of the UV-projected row

$$d_{\perp} = 2 - 2\epsilon, \overline{\text{MS}} \text{ — every step}$$

$$\int d^2 r \, e^{iq \cdot r} \frac{1}{\xi D} \frac{s \cdot P}{P^2 r^2} \left[\xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\bar{\xi}} \right], \text{ then } \int d\xi$$

0. Conventions

$D = 4 - 2\epsilon$ spacetime, so the transverse plane is $d_{\perp} = 2 - 2\epsilon$; $g^2 \rightarrow g^2 \mu^{2\epsilon}$, with one power of $\mu^{2\epsilon}$ accompanying the divergent transverse integral (the other multiplies a finite quantity and is $O(\epsilon)$). The $\overline{\text{MS}}$ scale is

$$\bar{\mu}^2 \equiv 4\pi\mu^2 e^{-\gamma_E}.$$

Kinematics unchanged: $r = z - x$, $s = y - x$, $P = x' - w'$, $\xi = p^+/k^+$, $\bar{\xi} = 1 - \xi$, $q = \bar{\xi}k$, and

$$D = (r - s)^2 + M^2, \quad M^2 = \frac{\xi}{\bar{\xi}} s^2, \quad D_0 \equiv D|_{r=0} = s^2 + M^2 = \frac{s^2}{\bar{\xi}},$$

$$C_{\text{UV}}(\xi) = \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}(\xi)}{2N_c}, \quad \ell_k = \log \frac{k^+}{\Lambda}.$$

Step 0. C_{UV} carries no $O(\epsilon)$. The instantaneous δ^{im} vertex supplies an explicit $d_{\perp}$ and the angular average supplies $1/d_{\perp}$; they cancel, $c_1 \langle S_1 \rangle = \frac{d_{\perp}}{d_{\perp}} \xi \bar{\xi} \frac{s \cdot P}{r^2}$. So the projected bracket is exact in any $d_{\perp}$, and nothing multiplying the $1/\epsilon$ pole is itself $O(\epsilon)$ — no extra finite term is generated by $\epsilon \times \frac{1}{\epsilon}$.

1. The transverse integral is the only thing to regulate

$C_{\text{UV}}(\xi)$ and $(s \cdot P)/P^2$ do not depend on r ; the whole r -dependence is $1/(r^2 D)$ and the phase. So the object to continue is

$$\mathcal{G}_{\epsilon} = \mu^{2\epsilon} \int d^{2-2\epsilon} r \frac{e^{iq \cdot r}}{r^2 D}.$$

2. Split the divergence off exactly

$$\frac{1}{r^2 D} = \frac{1}{r^2 D_0} + \frac{1}{r^2} \left(\frac{1}{D} - \frac{1}{D_0} \right), \quad \frac{1}{D} - \frac{1}{D_0} = \frac{D_0 - D}{D D_0} = \frac{2r \cdot s - r^2}{D D_0},$$

and the second bracket vanishes linearly at $r = 0$, so that piece is finite and may be taken at $\epsilon = 0$:

$$\mathcal{G}_{\epsilon} = \frac{1}{D_0} \mu^{2\epsilon} \int d^{2-2\epsilon} r \frac{e^{iq \cdot r}}{r^2} + \frac{1}{D_0} \left[2s \cdot \Psi - \Phi \right].$$

3. The d -dimensional Fourier transform

$$\int d^d r \frac{e^{iq \cdot r}}{(r^2)^a} = \frac{\pi^{d/2} 2^{d-2a} \Gamma(\frac{d}{2} - a)}{\Gamma(a)} (q^2)^{a-d/2},$$

at $a = 1$, $d = 2 - 2\epsilon$, times $\mu^{2\epsilon}$:

$$\mu^{2\epsilon} \int d^{2-2\epsilon} r \frac{e^{iq \cdot r}}{r^2} = \pi^{1-\epsilon} 2^{-2\epsilon} \Gamma(-\epsilon) \mu^{2\epsilon} (q^2)^{\epsilon} = \pi \Gamma(-\epsilon) \left(\frac{q^2}{4\pi\mu^2} \right)^{\epsilon}.$$

(Checked against the known $d = 3$ case, where the formula gives $2\pi^2/|q|$; §8.)

4. Expand in ϵ

$\Gamma(-\epsilon) = -\frac{1}{\epsilon} - \gamma_E + O(\epsilon)$ and $\left(\frac{q^2}{4\pi\mu^2} \right)^{\epsilon} = 1 + \epsilon \log \frac{q^2}{4\pi\mu^2} + O(\epsilon^2)$, so

$$\pi \Gamma(-\epsilon) \left(\frac{q^2}{4\pi\mu^2} \right)^{\epsilon} = -\pi \left[\frac{1}{\epsilon} + \gamma_E + \log \frac{q^2}{4\pi\mu^2} \right] + O(\epsilon) = -\pi \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{q^2}{\bar{\mu}^2} \right],$$

the γ_E and $\log 4\pi$ packaging into $\bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_E}$ exactly.

Sign of the pole. It is $-1/\epsilon$, not $+1/\epsilon$. A short-distance ($r \rightarrow 0$) singularity in coordinate space is regulated by $d_\perp > 2$, i.e. by $\epsilon < 0$ in the convention $d_\perp = 2 - 2\epsilon$ tied to $D = 4 - 2\epsilon$; the continuation returns $\Gamma(-\epsilon)$, whose pole is $-1/\epsilon$. For $\epsilon < 0$ this is $+\pi/|\epsilon| > 0$, matching the positive $\pi \log(1/m^2)$ of a mass regulator. In the opposite convention $d_\perp = 2 + 2\epsilon$ the same object reads $+1/\epsilon$.

5. $\overline{\text{MS}}$ subtraction

$$\begin{aligned} \mathcal{G}_\epsilon &= \frac{\pi}{D_0} \left[-\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\bar{\mu}^2}{q^2} + \tilde{I}(\xi) \right], & \mathcal{G}_\epsilon \Big|_{\overline{\text{MS}}} &= \frac{\pi}{D_0} \left[\log \frac{\bar{\mu}^2}{q^2} + \tilde{I}(\xi) \right], \\ \tilde{I}(\xi) &\equiv \frac{2 s \cdot \Psi - \Phi}{\pi}, & \Phi &= 2\pi e^{iq \cdot s} K_0(M|q|), \\ s \cdot \Psi &= \pi \int_0^1 d\alpha e^{i\bar{\alpha} q \cdot s} \left[\bar{\alpha} s^2 \frac{|q|}{\sqrt{\Delta}} K_1(\lambda) + i(q \cdot s) K_0(\lambda) \right], & \Delta &= \bar{\alpha}(\alpha s^2 + M^2), \lambda = |q|\sqrt{\Delta}. \end{aligned}$$

Equivalently, as a single convergent parameter integral, $\tilde{I} = \hat{I} + \log \frac{q^2 D_0 e^{2\gamma_E}}{4}$ with $\hat{I} = D_0 \int_0^1 dt [e^{itq \cdot s} |q| K_1(|q|\sqrt{\Delta_0})/\sqrt{\Delta_0} - \frac{1}{tD_0}]$, $\Delta_0 = t[(1-t)s^2 + M^2]$.

6. Back into the row — the $\bar{\xi}$ cancels

Since $\frac{1}{\bar{\xi}} \cdot \frac{1}{D_0} = \frac{\bar{\xi}}{\xi s^2} = \frac{1}{s^2}$,

$$\frac{1}{\bar{\xi}} \frac{s \cdot P}{P^2} C_{\text{UV}}(\xi) \mathcal{G}_\epsilon = \pi C_{\text{UV}}(\xi) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \left[-\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\bar{\mu}^2}{\xi^2 k^2} + \tilde{I}(\xi) \right],$$

using $q = \bar{\xi}k$. The transverse structure is a product of two Weizsäcker–Williams kernels.

7. The p^+ integral

$\log \frac{\bar{\mu}^2}{\xi^2 k^2} = \log \frac{\bar{\mu}^2}{k^2} + \log \frac{1}{\xi^2}$, and

$$\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi C_{\text{UV}}(\xi) = 2\ell_k - \frac{11}{6}, \quad \int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi C_{\text{UV}}(\xi) \log \frac{1}{\xi^2} = \ell_k^2 + \frac{\pi^2}{3} - \frac{67}{18}.$$

The second is exactly minus Eq. (C.10) of arXiv:1610.03453, since $\log \bar{\xi}^{-2} = -\log(\xi \bar{\xi}) + \log(\xi/\bar{\xi})$ and the second piece is odd under $\xi \leftrightarrow \bar{\xi}$ while C_{UV} is even. Both verified to 13 digits (§8).

8. Final result

Collecting the prefactor, $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4 k^+} \times \frac{\pi}{2\pi} = \frac{g^4}{128\pi^7 k^+} = \frac{\alpha_s^2}{8\pi^5 k^+}$, and pulling the N_c out of the colour bracket:

$$\frac{d^3 N_2^{\text{NLO}}}{d^3 k} = \frac{\alpha_s^2(\bar{\mu}) N_c}{8\pi^5 k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} [\mathcal{P} + \mathcal{F}]$$

$$\times [U^{ab'}(x') - U^{ab'}(w')] [U^{af}(y) - U^{af}(x)] \rho^{b'}(x') \rho^f(y),$$

$$\mathcal{P} = -\frac{1}{\epsilon_{\text{UV}}} \left(2\ell_k - \frac{11}{6} \right) \xrightarrow{\overline{\text{MS}}} 0$$

$$\mathcal{F} = \left(2\ell_k - \frac{11}{6} \right) \log \frac{\bar{\mu}^2}{k^2} + \ell_k^2 + \frac{\pi^2}{3} - \frac{67}{18} + \int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi C_{\text{UV}}(\xi) \tilde{I}(\xi)$$

Three readings.

- **The pole is the coupling renormalisation.** $\mathcal{P} \times 2N_c = -\frac{1}{\epsilon} (4N_c \ell_k - \frac{11N_c}{3}) = -\frac{1}{\epsilon} (4N_c \ell_k - b)$ at $N_f = 0$: the $\frac{11}{6}$ is the b -function coefficient, and this is what cancels against the $-2/\epsilon$ of the gluon self-energy counterterm. In $\overline{\text{MS}}$ it is simply removed, together with $-\gamma_E + \log 4\pi$, which is the content of $\bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_E}$.
- **The scale in the logarithm is k^2 , not the dipole size.** After the $\bar{\xi}$ cancellation of §6 the only scale the projected integrand retains is $q = \bar{\xi}k$, so $\overline{\text{MS}}$ delivers $\log(\bar{\mu}^2/k^2)$ and the $\bar{\xi}$ -dependence becomes the $\ell_k^2 + \frac{\pi^2}{3} - \frac{67}{18}$. Trading it for $\log(\bar{\mu}^2(y-x)^2)$ is a rescheming, which moves $\log(k^2(y-x)^2)$ between the two and is not done here.
- **What is left over.** $\tilde{I}(\xi)$ is finite and regulator-independent — the Bessel remainder of §5, unaffected by the continuation. Its ξ -integral is not elementary; its endpoints give $e^{ik \cdot (y-x)} \frac{1}{2} \ell_k^2 + c_0 \ell_k + c_1$ as before.

9. Checks

```
A. the d-dimensional transform, formula checked where the answer is known (d=3, a=1)
pi^(d/2) 2^(d-2a) Gamma((d-2a)/2)/Gamma(a) = 19.739208802179
known result int d^3x e^(iqx)/x^2 = 2 pi^2/|q| , 2 pi^2 = 19.739208802179

B. the epsilon expansion: pi Gamma(-eps) (q^2/(4 pi mu^2))^eps + pi/eps -> pi log(mubar^2/q^2)
q^2 = 2.7, mu^2 = 1.3, mubar^2 = 4 pi mu^2 e^-gamma = 9.17216034208
eps= 1e-3: Gamma-form + pi/eps = 3.8369896778214 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-4: Gamma-form + pi/eps = 3.8414265852134 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-5: Gamma-form + pi/eps = 3.8418705304043 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-6: Gamma-form + pi/eps = 3.8419149274703 pi log(mubar^2/q^2) = 3.8419198605062
eps= 1e-7: Gamma-form + pi/eps = 3.8419193672023 pi log(mubar^2/q^2) = 3.8419198605062

C. the p+ moments (lambda = 1e-9, l_k = log(1/lambda))
int C_UV = 39.613198342559 2 l_k - 11/6 = 39.613198340559
int C_UV log(1/xib^2) = 429.02139290167 l_k^2 + pi^2/3 - 67/18 = 429.02139286022
int C_UV log(xi/xib) = -7.6498852e-22 (odd under xi <-> xibar, vanishes)
int C_UV log(xi xib) = -429.02139290167 [paper (C.10)]; sum with the row above: -7.6498848e-22
```

zint/src/run_dimreg.py — the numbers above.